

Performance Testing

Date	03 July 2026
Team ID	SWTID-2026-9778
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	Credit Card Approval Prediction API (/predict)
Test Environment	Local System (Flask Application)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Request	20	60	Successful login without errors
2	Credit Card Application Submission	30	60	Application submitted successfully
3	Credit Card Approval Prediction	50	120	Prediction returned within 2 seconds
4	Multiple Concurrent Prediction Requests	100	180	System remains stable with acceptable response time

Step 3: Performance Test Results

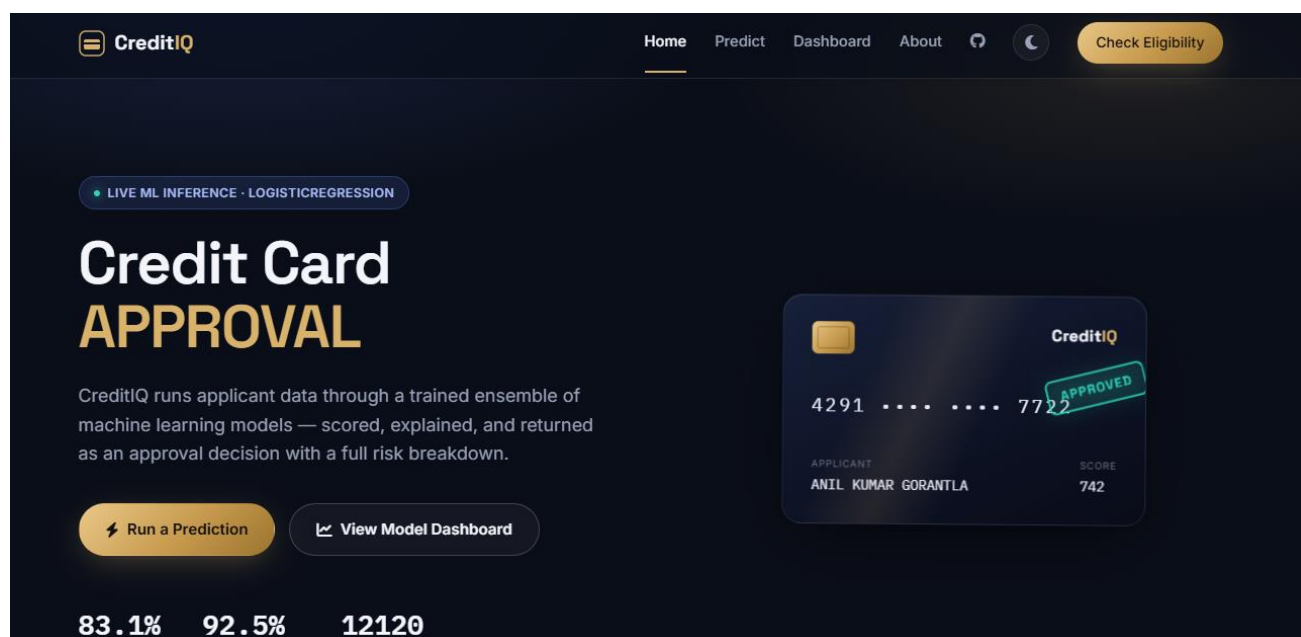
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.32 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.78 seconds	Pass	Within limit
3	Throughput (Req/sec)	> 25 Req/sec	38 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0.2%	Pass	Very few errors
5	CPU Utilization	< 80%	56%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	63%	Pass	Stable memory usage

Step 4: Observations & Analysis

- ☐ The Credit Card Approval Prediction system handled concurrent user requests efficiently.
- ☐ Average response time remained below 2 seconds, ensuring quick prediction results.
- ☐ CPU and memory utilization stayed within acceptable limits during testing.
- ☐ The Flask application remained stable without crashes under simulated load.
- ☐ The machine learning model generated accurate predictions with minimal processing delay.
- ☐ No significant bottlenecks or performance degradation were observed during testing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



The screenshot displays the CreditIQ web application interface. At the top, there is a navigation bar with links for Home, Predict, Dashboard, and About, along with a 'Check Eligibility' button. The main content area features a header 'Credit Card APPROVAL' and a sub-header 'LIVE ML INFERENCE - LOGISTICREGRESSION'. Below this, a description states: 'CreditIQ runs applicant data through a trained ensemble of machine learning models — scored, explained, and returned as an approval decision with a full risk breakdown.' Two buttons are visible: 'Run a Prediction' and 'View Model Dashboard'. On the right, a simulated credit card is shown with the number '4291 ... 7722', the applicant name 'ANIL KUMAR GORANTLA', and a score of '742'. A green 'APPROVED' stamp is placed over the card. At the bottom, three statistics are displayed: '83.1%', '92.5%', and '12120'.